



Quantum Computing at Infleqtion: Toward Tightly Integrated Quantum-HPC Systems

openQSE Community Talk

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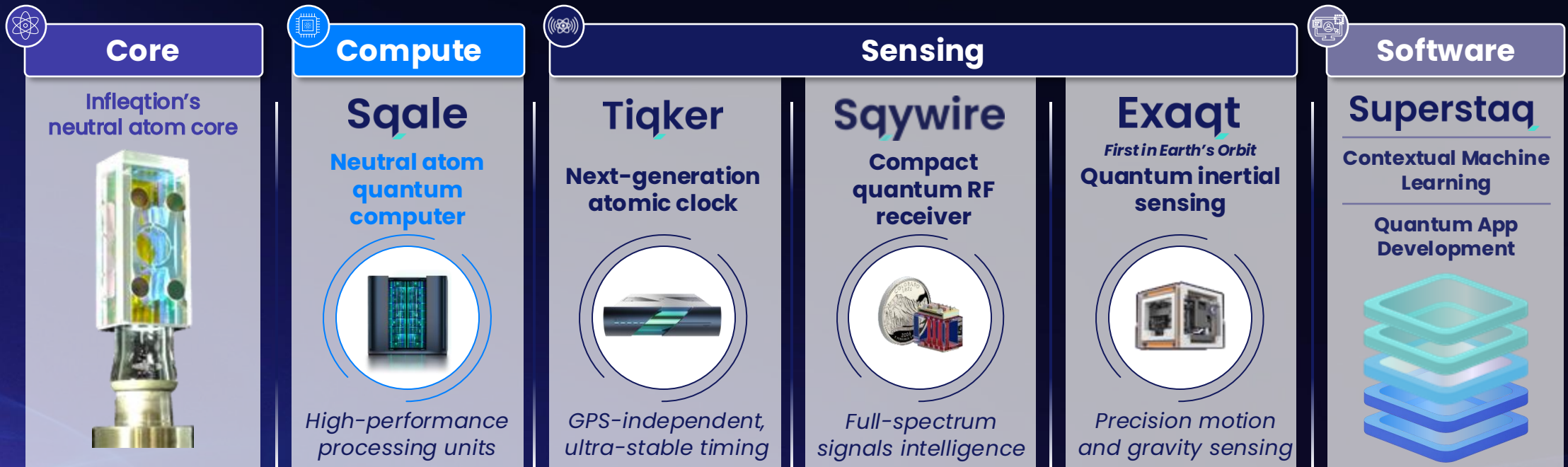


August 17, 2026

What we do



We design, build, and sell **quantum computers, precision sensors, and software** to governments, corporations, and research institutions



The only full-spectrum quantum company

Growing, Global Presence

World-Class Development Centers & Research Partners



International
Space Station



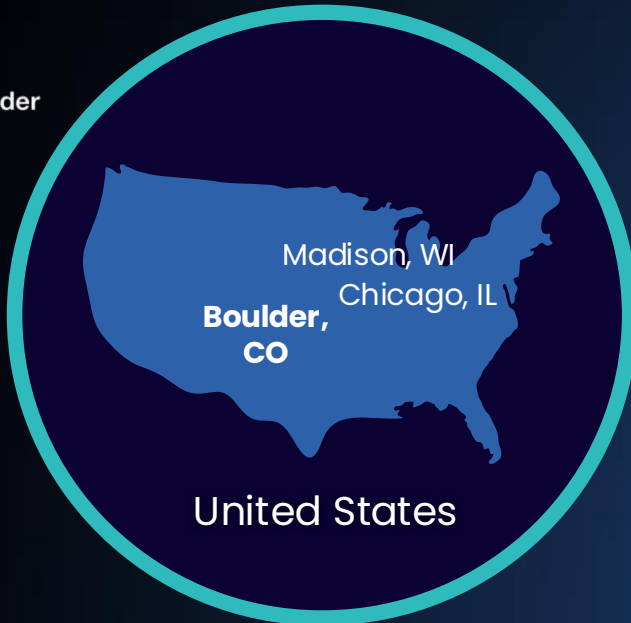
Melbourne,
Australia



Okazaki,
Japan



Oxford,
United Kingdom



United States

 University of Colorado **Boulder**

 **WISCONSIN**
UNIVERSITY OF WISCONSIN-MADISON

 THE UNIVERSITY OF
CHICAGO

 Massachusetts
Institute of
Technology

UC SANTA BARBARA

Our products in use



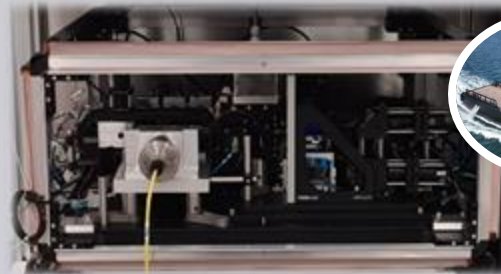
On the **ground**, in the **sea**, in the **sky**, and in **space**

First quantum computer installation at the National Quantum Computing Centre



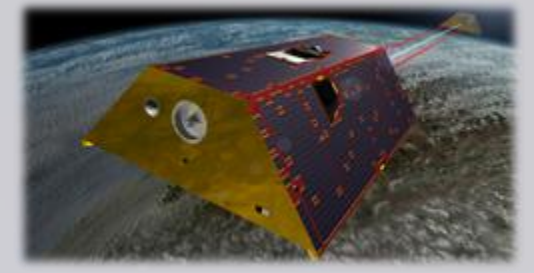
Only foreign company to be selected to participate in Japan Moonshot R&D program

Sea trial of hybrid navigation systems on Royal Navy's experimental vessel



First-ever successful demonstration flight of a commercial optical clock

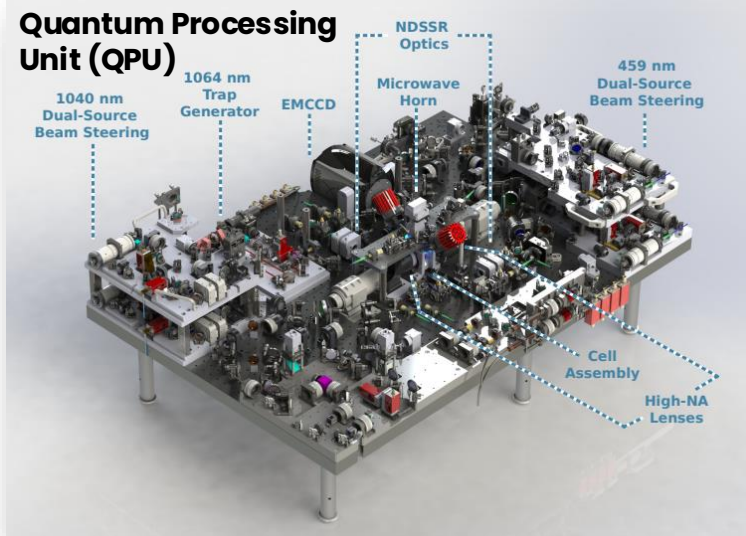
Pilot program for putting quantum experiments and sensors in space



NASA Quantum Gravity Gradiometer Pathfinder Program

Quantum Computing @ Infleqtion

Quantum Processing Unit (QPU)



*The heart of all
our products*



*Vacuum cell
with a cesium
atom array
inside*

*Atoms are
controlled
using external
lasers*

- ❖ **Single-species** machines in Colorado, UK, and Japan
- ❖ **Dual-species** (Rb and Cs) in Wisconsin
- ❖ Bringing our machine to the **Illinois Quantum and Microelectronics Park**

Neutral atoms = nature's perfect qubits



Natural advantage

Ideal uniformity, stability, and long-lasting performance



Commercially scalable

Can be arranged in large arrays using optical tweezers



Longer coherence times

Surpassing other modalities in qubit growth and control



Exceptional control

Less impacted by "noise" from the outside world

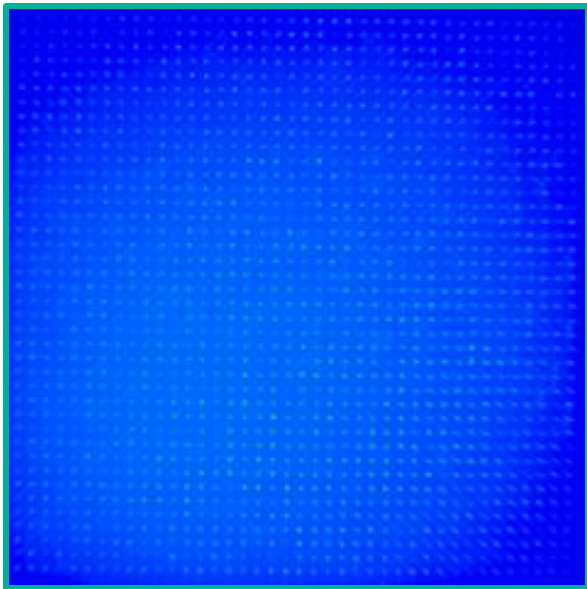


Universal connectivity

Any-to-any qubit coupling enables optimal algorithm execution

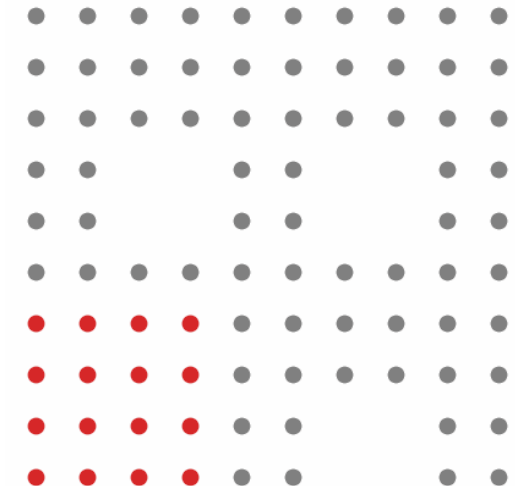
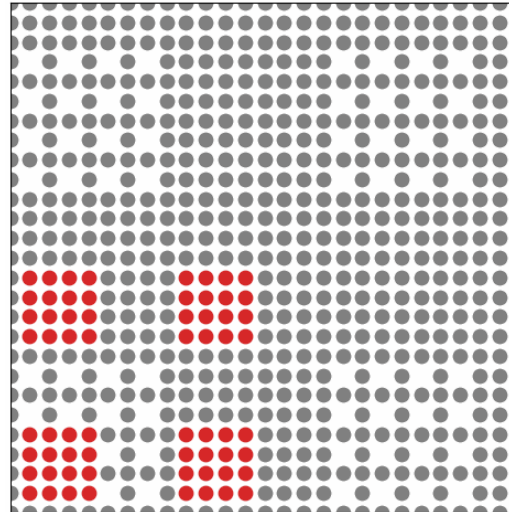
Feature Spotlight: Dynamic Reconfiguration

Highly Scalable Array



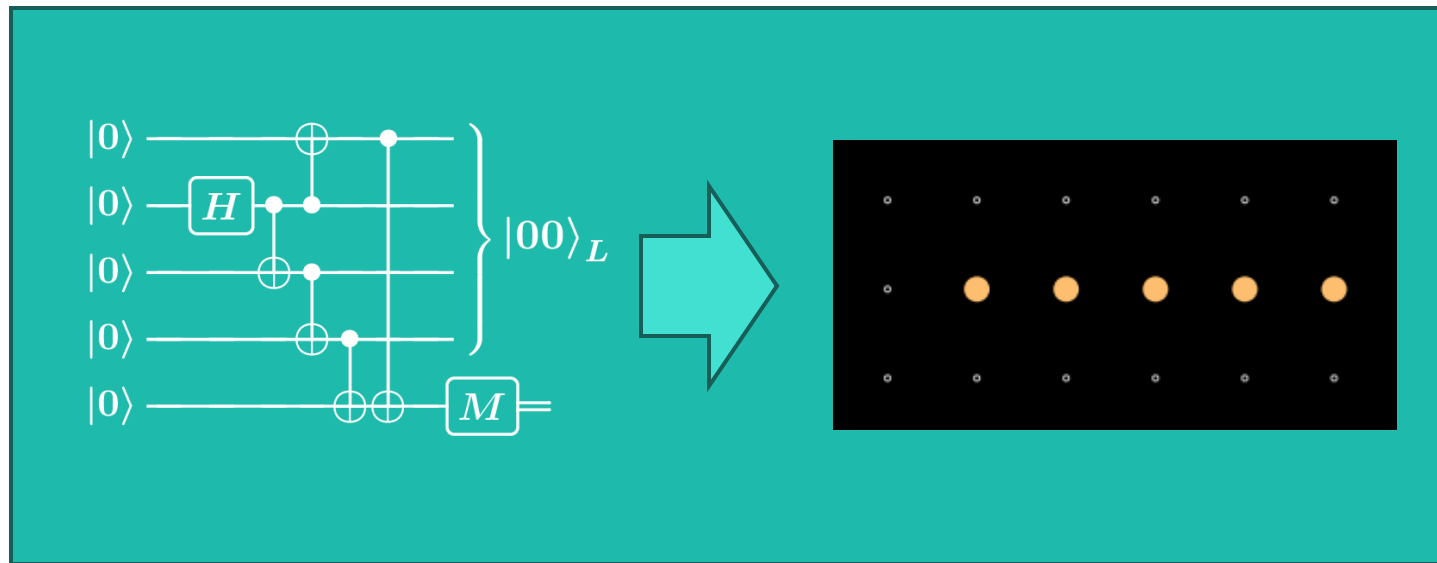
Single Cesium Atoms in
 $40 \times 40 = 1,600$ site array
(averaged image)

Dynamic atom array configuration



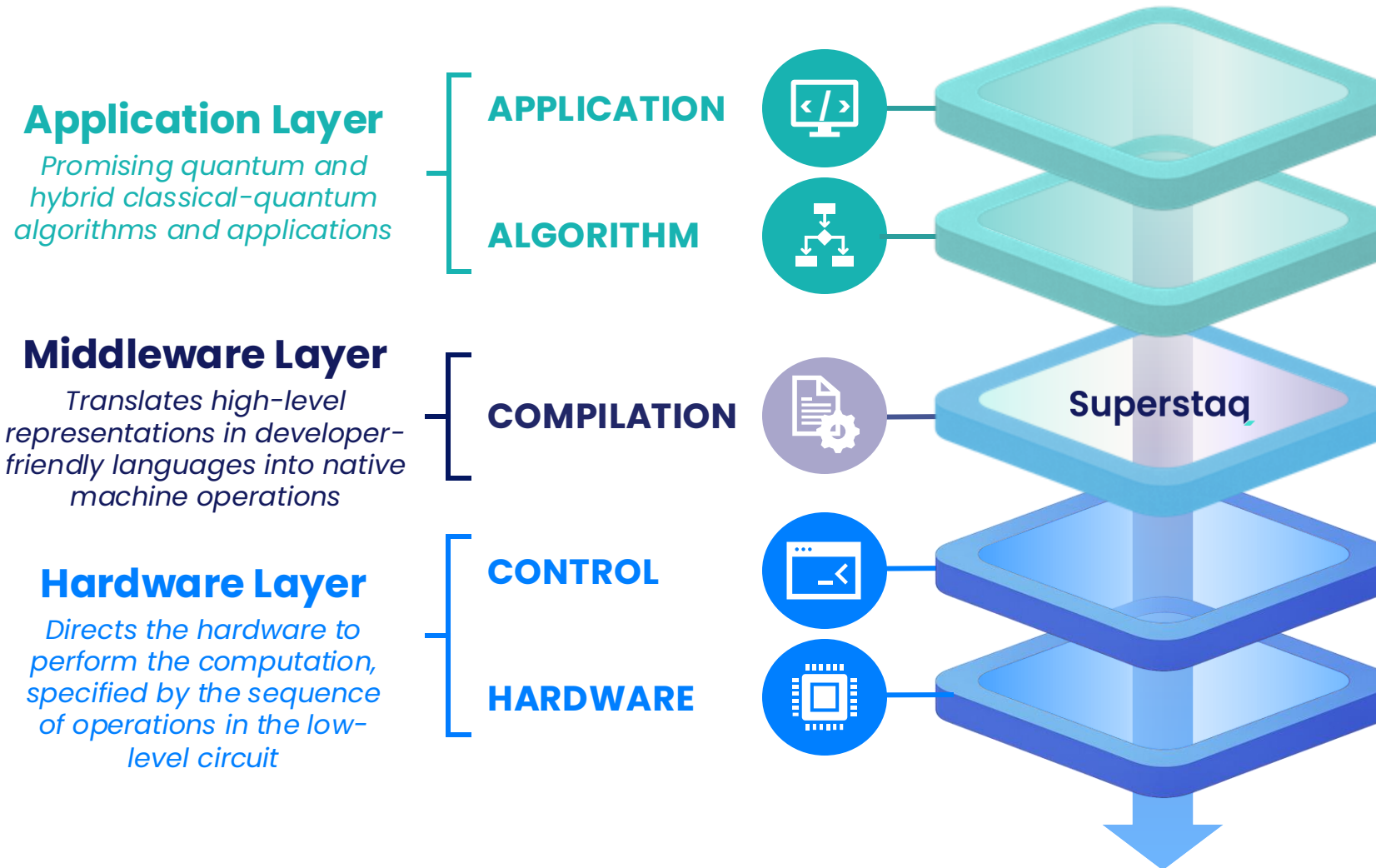
Dynamic Reconfiguration and FTQC

Fault-tolerant $|00\rangle$



Dynamic Reconfiguration supports all-to-all configuration and coherent feedback

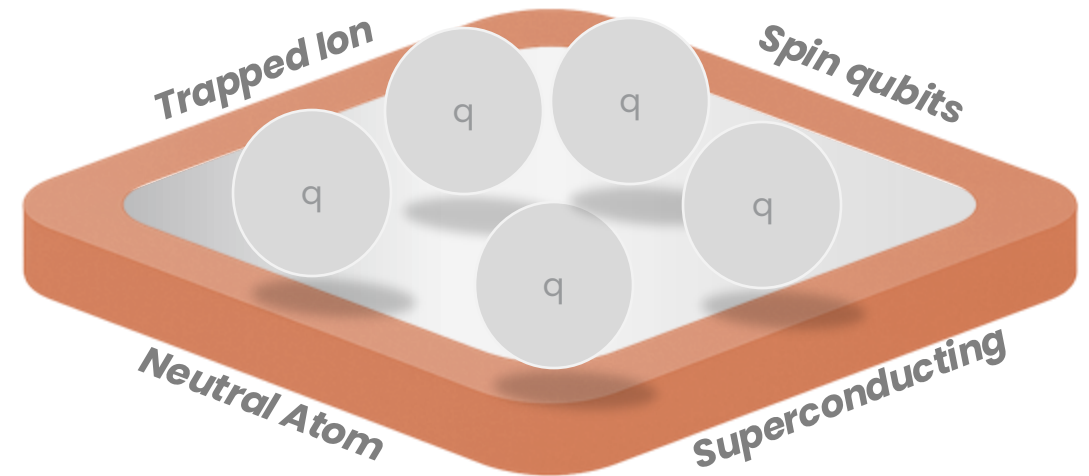
Infleqtion's Quantum Computing Stack



Superstaq

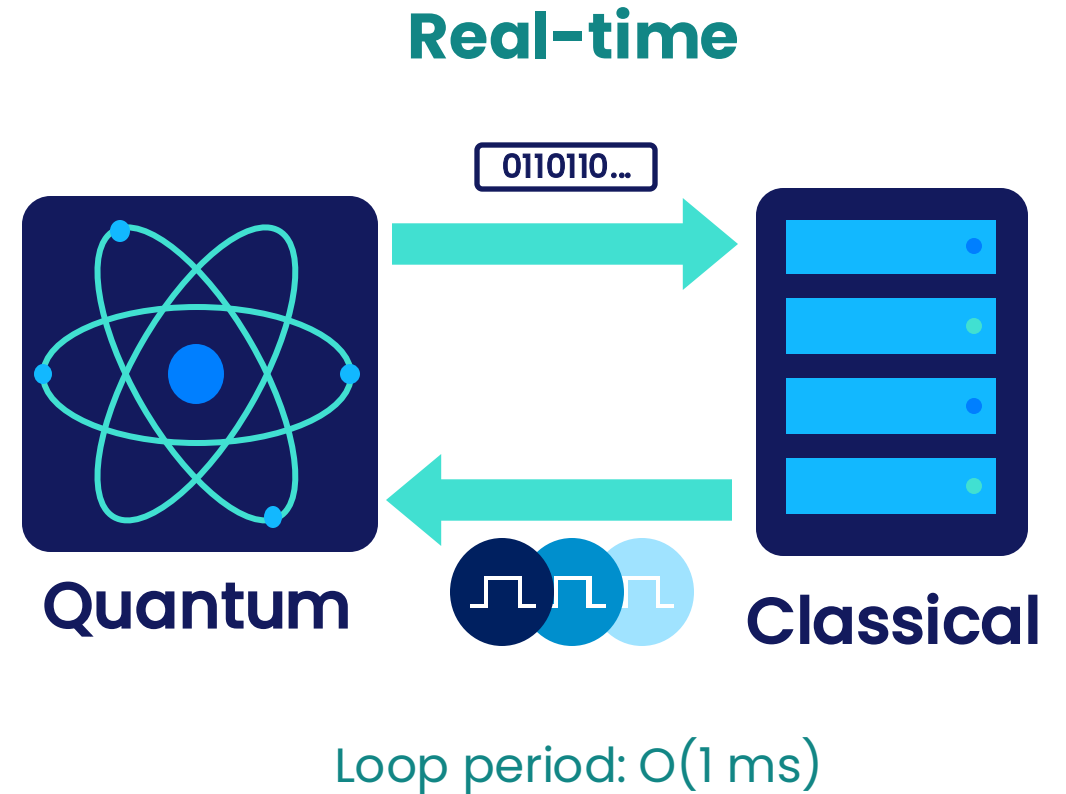
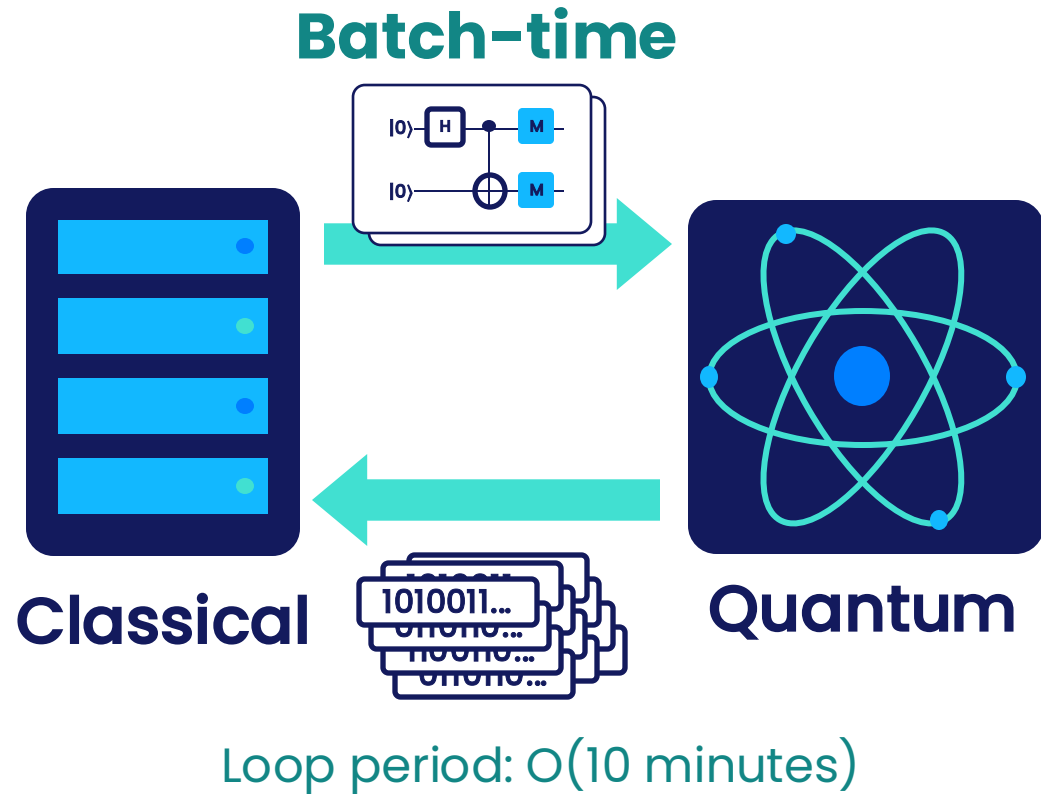
Hardware-agnostic software platform operational across modalities

- Enables quantum programming in any source language using any standard quantum computing developer tool
- Optimized compiler backends for Infleqtion and many third-party quantum computers and simulators

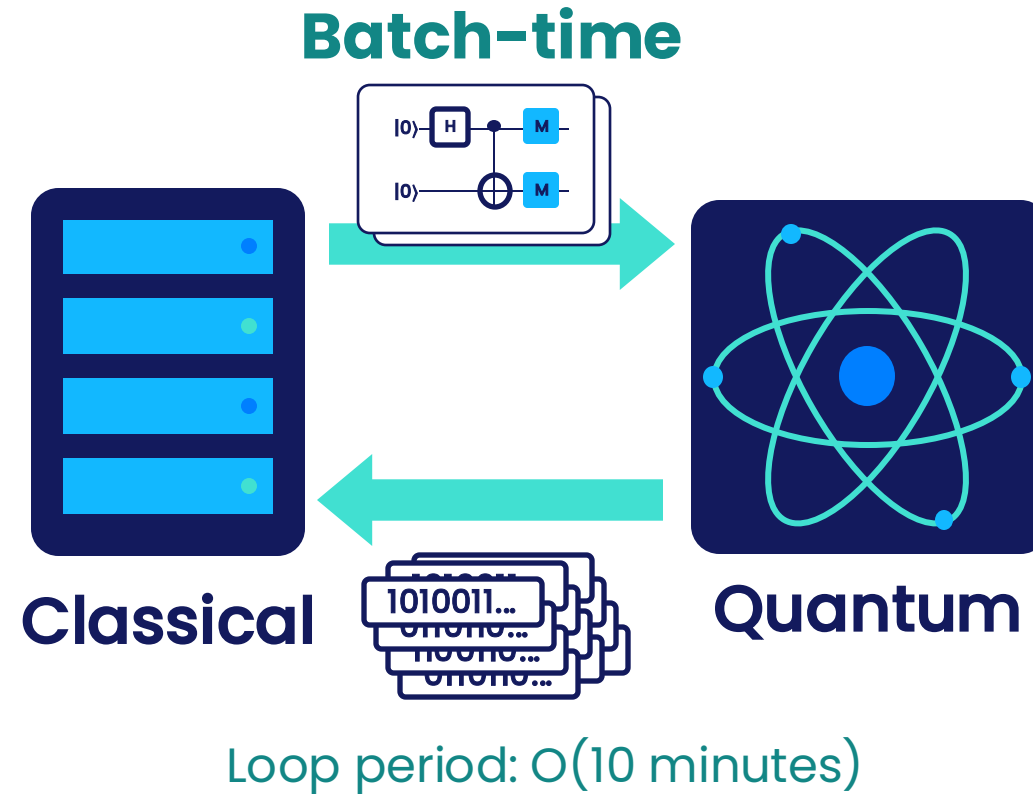


<https://superstaq.readthedocs.io>

Hybrid Classical-Quantum Workflows

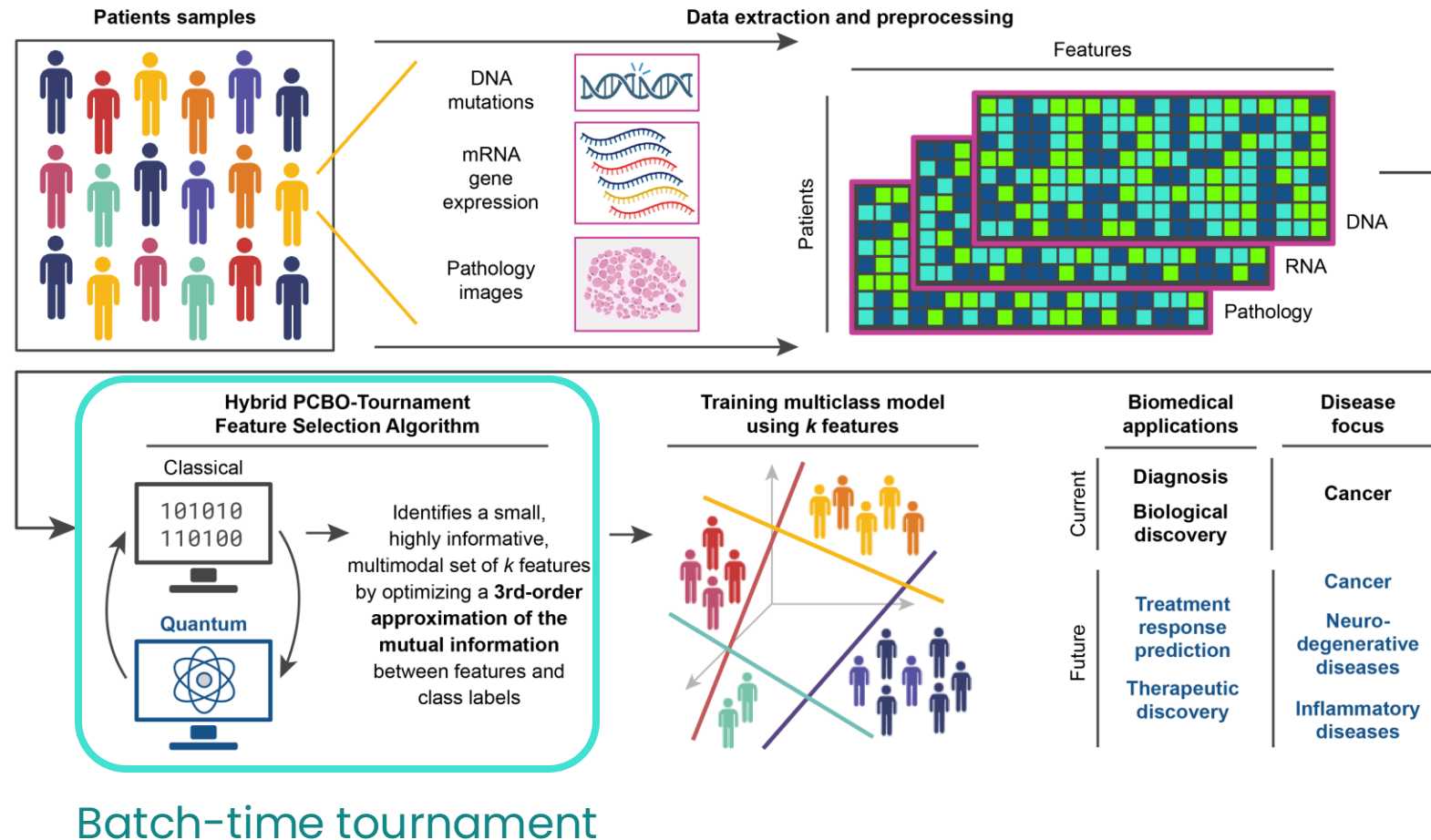


Batch-time Classical-Quantum Workflows

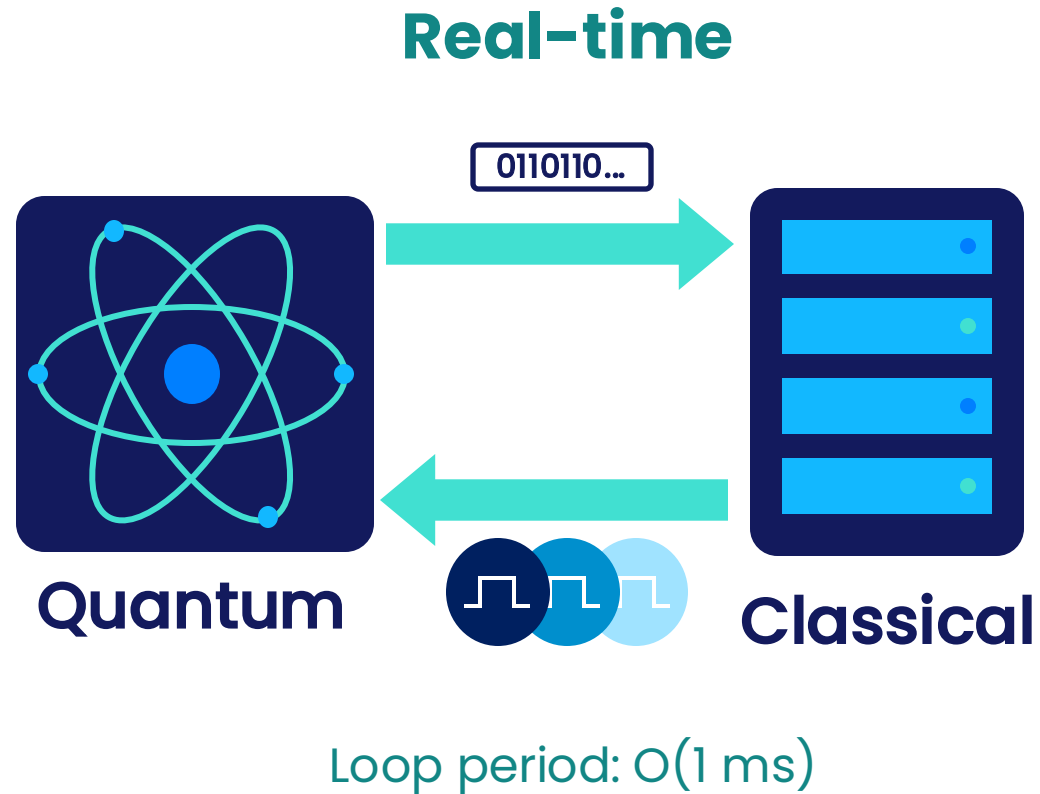


Batch-time application: Precision Oncology

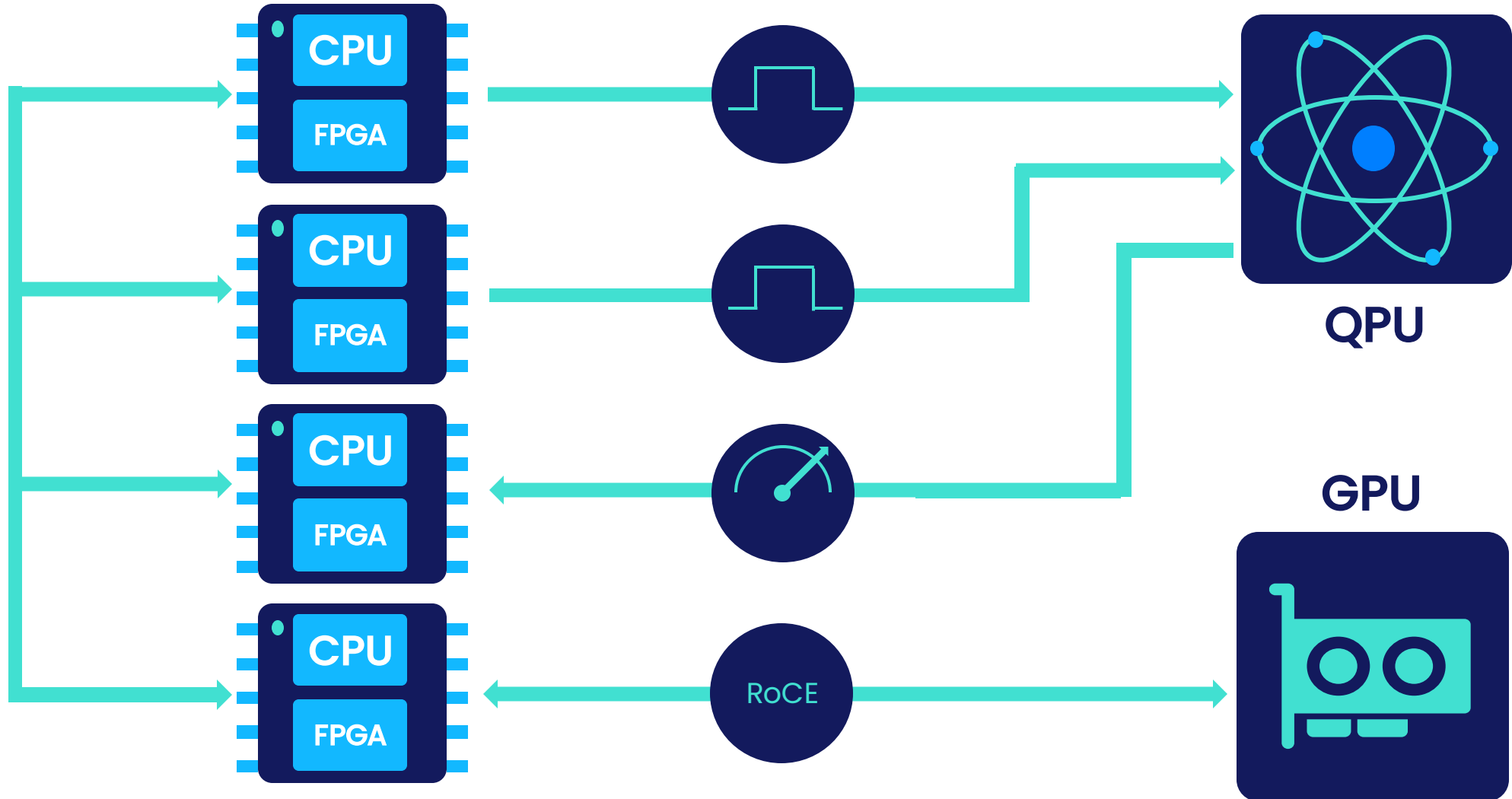
Wellcome Leap Q4Bio: Partnership with UChicago + MIT



Realtime Classical-Quantum Workflows



Realtime Hardware



Heterogeneous Kernels

```
H(q1)
CNOT(q1, q2)
m1 = measure(q1, q2)
result = decode(m1)
```

Quantum Instructions

Compilation

```
play_pulse(h_q1)
play_pulse(cnot_q1_q2)
m1 = measure(q1_q2_reg)
result = dispatch_fn(
    "decode", m1
)
```

RTCPU

```
result_t decode(m1 *uint8_t){
    // GPU decoding
    // routine goes here
}
```

GPU

Classical Instructions

Thank you for listening!

Questions?
Comments?